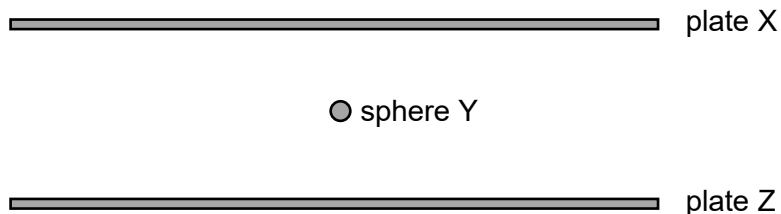


- 22** The diagram shows two charged horizontal metal plates X and Z with a potential difference set up between them. A small sphere Y with charge Q is suspended and remain stationary at the midpoint between the plates.



What will happen to sphere Y when its charge increases?

- A** Sphere Y will accelerate downwards and hit plate Z
- B** Sphere Y will move down nearer to plate Z and remain stationary.
- C** Sphere Y will accelerate upwards and hit plate X.
- D** Sphere Y will move up nearer to plate X and remain stationary.

Ans: C

Y is stationary as its weight balanced the electric force

$$mg = qE$$

when the charge increases, electric force increase and sphere Y will accelerate upwards due to a resultant force.

- 23** Four parallel current carrying conductors are placed vertically at the corners of a square WXYZ